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SOLAR POWER FORECASTING USING ARTIFICIAL INTELLIGENCE

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ABSTRACT

A mathematical model has been developed that predicts the electricity production of photovoltaic power plant based on weather forecast using neural networks and machine learning. We have also developed an internet application that includes this model and can be used online. We successfully tested the application during the years 2024-2025, and the results show that it can predict electricity production with high accuracy for several days into the future and compare expected and actual production retrospectively. The accuracy of the model naturally depends on the accuracy of the weather forecast. The model uses data from the Open-Meteo service, which has high accuracy for one day in advance and good accuracy for three days in advance. The model has direct practical significance for photovoltaic power plant operators, especially in today's spot price environment in Europe.

Key words

Photovoltaics, PV power plant, neural network, machine learning, mathematical model.

INTRODUCTION

Renewable energy sources (RES) are permanently gaining importance. This is related to climate change and the effort to reduce CO_2 emissions. Diversifying energy supplies through RES ensures sustainability. However, the ever-growing interest in increasing the share of RES requires the integration and proper management of systems that use multiple technologies (for example photovoltaics, solar thermal energy, wind energy, hydropower). To strengthen these management processes, short- and medium-term forecasting of possible energy supplies from individual RES is very important. Currently, there are several comprehensive methods based on artificial intelligence that can be used to forecast energy production and next parameters (Ren et al, 2026, ElRobrini et al, 2024, Khan et al, 2022). However, there is no available methodology for determining the actual energy performance of existing photovoltaic systems.

Currently, there is a solar boom mainly in Central Europe corresponding with the increase in energy prices. In the Czech Republic, around 150,000 small PV power plants were installed on residential buildings with a nominal output of around 10 kWp at the beginning of 2025. The number of all PV power plant installations was around 170,000. The total installed nominal output of all PV power plants in the Czech Republic was around 3,900 MWp. Agrovoltaic systems are also gaining importance (Libra et al, 2024, Poulek et al, 2024), which allow for dual use of land, which is also advantageous from the point of view of sustainable development.

Photovoltaic (PV) power plants have their advantages and disadvantages. The advantages include relatively simple installation and minimal maintenance and overhead. The disadvantages include low density of installed power per power plant area and, above all, instability of output power, which depends mainly on the weather. However, there are also influences independent of weather, such as solar eclipses (Libra et al, 2016). This is a rare but very significant influence. A good forecast of electricity production based on weather forecast can help PV power plant operators optimize financial profit, because energy prices in Europe are currently spot prices and change every 15 minutes according to the current situation in the distribution network.

A similar issue of short-term output power predictions was addressed, for example, in (Wang et al, 2024) and one-day-ahead predictions were addressed in (Wang et al, 2020). The work (Sadowska et al,

2024) discusses the creation and validation of a real-world energy model of PV systems based on detailed data from a 1 MWp photovoltaic farm in the Lublin Voivodeship (Poland). The difference between actual and predicted production was in the range of 15÷16%. The work (Su et al, 2025) uses neural networks and machine learning for ultra-short-term predictions and achieves similar accuracy.

MATERIALS AND METHODS

The prediction model was created for a photovoltaic power plant located in the village of Kunžak in the Czech Republic). It has a nominal output of 1,293 kWp. It consists of 5,625 photovoltaic panels. Each panel has a nominal output of 230 Wp with a permissible tolerance of ±5%. The panels had an initial efficiency of 14.1% at the time of installation (2010).

Data from the ABA station covering four years (2020–2024) at hourly intervals were used to train and test PV power output models. The dataset included historical PV power and meteorological variables (global solar radiation, panel temperature), supplemented with weather forecasts from OpenMeteo, such as atmospheric pressure, air humidity, wind speed and direction at 10 m, cloud cover, precipitation, diffuse and direct solar radiation, shortwave radiation, dew point, ozone concentration, PM10, and ambient temperature. Additional features were computed using the pvlib library, including the sun's zenith and azimuth and the maximum global solar radiation. Temporal features capturing daily and seasonal cycles (sin_hour, cos_hour, sin_day_of_year, cos_day_of_year) were included, along with wind components (wind_u, wind_v) derived from wind speed and direction to represent east-west and north-south flow. Extensive preprocessing was performed, including correlation analysis, Yeo-Johnson transformation, standardization, and Min-Max scaling. Experiments confirmed that the combination of these features contributed most to model accuracy.

Fig. 1 shows the linear correlation of the resulting features used for the model that contribute most to the output power of the PV power plant.

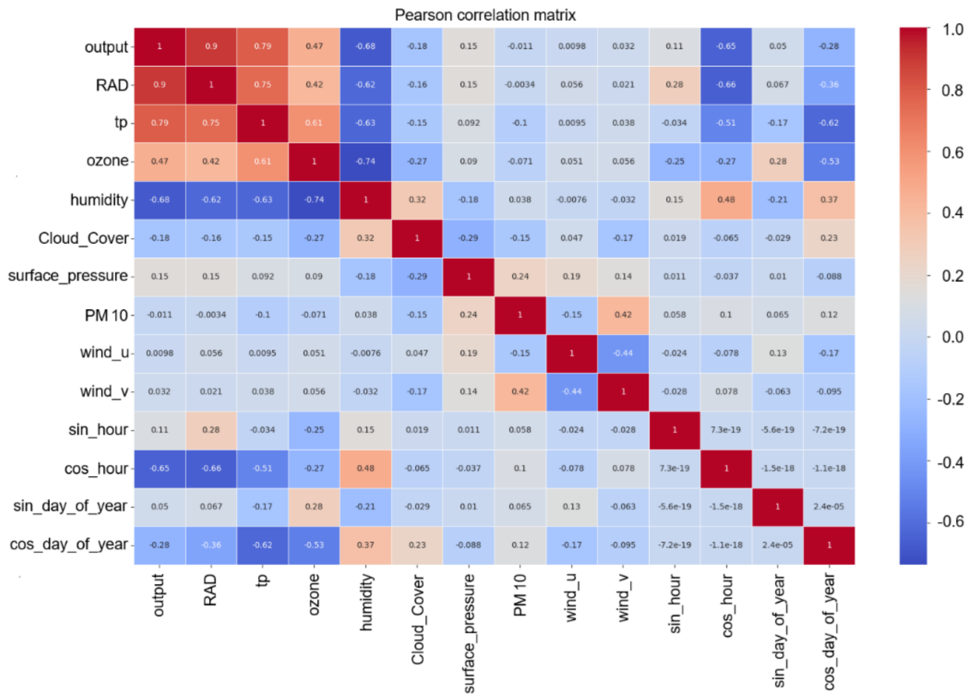

Fig. 1 Correlation matrix

Subsequently, the dataset was divided into a training (80%) and a testing (20%) set, with the testing set containing 3,507 records.

The final models developed are:

- Model based on historical averages of output power for each hour and day of the year
- Physical model using the basic formula for calculating the power of a PV panel
- Neural network-based model with weather forecasting

The metrics sMAPE, MAE, MBE and the coefficient of determination R^2 are used to evaluate their predictions. These metrics provide a comprehensive view on the accuracy of the models from different perspectives.

Model based on historical averages of power for each hour and day of the year

The reference model, which predicts output power based on historical averages of power for a given hour and day of the year, without taking into account current meteorological conditions, provides a certain level of prediction of PV power, but its accuracy is significantly lower compared to the reference physical models listed below. The model shows high errors: the relative error sMAPE reaches 80%, the absolute average error MAE is 77.78 kW, the model systematically underestimates predictions with MBE -22.81 kW and the coefficient of determination R^2 is 0.64, which indicates a significantly worse ability of the model to explain the variability of actual power values.

Physical model using the basic formula for calculating the power of a PV panel

For the reference physical model, the formula from Eq. 1 was used to calculate the predicted output power of the PV power plant. Based on this equation, the predicted power values for individual hours were determined. The value of 35° was set for the angle β. It corresponds with the inclination of the PV panels. The resulting metrics show that the model shows a relatively high error rate. The relative error sMAPE reaches 75.1%, the absolute error MAE is 34.02 kW, and the negative value MBE -28.46 kW confirms that the model systematically underestimates the output power of the PV power plant. Nevertheless, the coefficient of determination R^2 at the level of 0.93 indicates that the model can capture a significant part of the variability of the actual power and provides a relatively accurate approximation.

$$P_{Teor} = E_{Total} * \eta * S * \cos\beta \tag{1}$$

P_{Theor} is the output power of the PV panel (W),
E_{Total} is the total solar radiation (W/m²) perpendicularly on the earth's surface,
η is the efficiency of the PV panel (-),
S is the area of the PV panel (m²),
β is the angle of inclination of the PV panel to the horizon (º).

A model based on neural networks with the inclusion of weather forecasting

Many experiments were performed with different input features and model architectures. To ensure consistency and replicability of the results, the same initial random number generator (seed) value was used in all experiments, ensuring consistent initialization of the model weights.

The most promising model architecture turned out to be a combination of a convolutional layer and an LSTM, which effectively captures both short-term and long-term patterns in the data. The convolutional layer extracts local patterns and short-term correlations between input variables, while the LSTM layer models long-term dependencies and seasonal cycles.

The input layer of the model expects a tensor with dimensions of 24 time steps × 26 input features. These include current measurements such as PV power output, global solar radiation (RAD), panel temperature (tp), cloud cover (Cloud_Cover), air humidity (humidity), dust particles (PM10), ozone (Ozone), surface atmospheric pressure (surface_pressure), wind components (wind_u, wind_v), and

time-related variables (sin_day_of_year, cos_day_of_year, sin_hour, cos_hour), as well as predicted meteorological features for the next 24 hours. The predicted features include ambient temperature (T2M), solar radiation (RAD), humidity, surface pressure, cloud cover, wind components (wind_u, wind_v), ozone concentration, dust particles (PM10), and the same time variables (sin_hour, cos_hour, sin_day_of_year, cos_day_of_year).
Keras Tuner with Bayesian optimization was used to find the optimal hyperparameters. Training was performed using cross-validation and a growing window; the number of epochs was set to 40 for the first run and 50 for the next. The optimized model includes a convolutional layer with 96 filters (kernel 2, ReLU activation, same padding) and an LSTM layer with 2816 units and a recurrent dropout of 10%. The output Dense layer corresponds to a 24-hour prediction horizon and uses a linear activation function. The other hyperparameters – ADAM optimizer, MAE loss function, learning rate 0.0005 and batch 64 – were taken from the previous model and proved to be successful.
Model achieves very low sMAPE values, which indicates high accuracy. The maximum MBE is only around ±7.28 kW, which is a very low value considering the total PV power. The coefficient of determination R^2 is stable at the value 1 and MAE values in the range of 9.8 to 18.3 kW indicate stable power prediction even with increasing time horizon.

RESULTS

We also tried to input the predicted power and the power predicted using the pvlib library into the model, but the results were almost identical, so they were not included in the resulting model.
Tab. 1 clearly shows the resulting evaluation metrics for all created models. The worst results are achieved by the model that predicts the power as the average historical value of a given hour in all previous years, which leads to an average sMAPE of 80%. The high error rate of this approach is expected because the model does not take into account current meteorological conditions and changes in the long-term trend of PV power.
The physical model calculates its predictions based on the physical Eq. 1 describing the relationship for calculating the output power of PV panels based on the intensity of incident solar radiation, the inclination of PV panels, their efficiency and the size of the active area. The model provides better results than historical averages, but still shows a relatively high error: sMAPE is 75.1%, the average MBE is −28.46 kW and the mean absolute error is 34.02 kW.
Higher accuracy of power predictions is achieved by the physical model, which replaces the constant value of cos(β) in Eq.7 with the correction coefficient IAM. This takes into account the actual angle of incidence of solar radiation and its effect on the amount of energy absorbed by the photovoltaic panels. The model shows overall better metrics: sMAPE 56.6%, MAE 17.28 kW and significantly reduces the systematic underestimation of predictions to -6.43 kW.
Models using neural networks with appropriately selected and processed inputs achieve completely different results. The most accurate prediction is provided by a model using a neural network, which, in addition to historical data, includes a weather forecast for the given predicted hours of power. This model achieves an average sMAPE of only 12.38% over the entire 24-hour prediction horizon, an average MBE of 1.96 kW, and the coefficient of determination R^2 of 0.98, which means an almost perfect match of the predicted values with the actual data. This result shows that the combination of historical data with current meteorological conditions allows the model to achieve the highest possible accuracy (of course, it depends on the accuracy of the weather forecast).
Overall results confirm that neural networks, especially in combination with weather forecasts, are the most suitable method for short-term prediction of the PV power plant output power. On the contrary, traditional methods, such as physical models without additional corrections or simple historical averages, show significantly worse results.

Tab. 1 Overview of evaluation metrics for all models.

Model	Time horizon	Average sMAPE (%)	Average MBE (kW)	Average R^2	Average MAE (kW)
Physical model	1 hour	75,1	-28,46	0,93	34,02
Historical averages	-	80	-22,81	0,64	77,78
Neural networks – with weather forecasting	24 hours	12,38	1,96	0,98	15,59

DISCUSSION

The experimental results clearly show that models based on neural networks achieve the highest accuracy in predicting the performance of photovoltaic power plants. The most accurate model, which includes weather forecasts in addition to historical data, achieves an average sMAPE error of 12.38%, which is significantly lower than that of physical models or the historical average data model. Moreover, its coefficient of determination $R^2 = 0.98$ confirms its ability to explain almost all the variability of the actual performance values.

Another important finding is the importance of selecting and transforming input features. The use of time-dependent cyclical variables and wind direction components contributed to higher prediction accuracy. It was also shown that the extent of the prediction horizon significantly affects accuracy - with increasing distance in time, the error rate increases, although a high-quality meteorological forecast significantly mitigates this effect.

The high-quality prediction of the output power of the PV power plant is also important from an economic perspective, as it can lead to more accurate estimation of spot electricity prices and optimization of electricity supply to the distribution grid.

CONCLUSION

In this work, a mathematical model for short-term prediction of photovoltaic power plant power. The model is created and tested based on neural networks combining convolutional (Conv1D) and LSTM layers. The highest accuracy was achieved by the model integrating meteorological forecast, which shows an average sMAPE of 12.38%, MAE of 15.59 kW and coefficient of determination $R^2 = 0.98$. The model was implemented in TensorFlow, trained in the Google Colaboratory environment and deployed in an online application on the Hugging Face platform (Hampejs, 2025), where it is freely accessible to the public for performance forecasting up to five days in advance and for retrospective evaluation as well.

The work compared various approaches to predicting PV power output – from simple models using historical data averages, through physical calculations, to advanced neural networks with various input combinations. The results showed that classical methods without taking into account current conditions significantly lag behind machine learning-based models.

Further improvement of prediction accuracy could be achieved by training the model on shorter time intervals (e.g. 15 minutes), combining physical and data models (ensemble approach), and especially using more accurate meteorological inputs. Forecast data from OpenMeteo show significant deviations in some cases; replacing them with locally calibrated data could significantly improve model performance.

The obtained results have direct practical application in the field of operational planning and optimization of PV power plant output power and form a solid basis for the further development of prediction tools in the field of renewable energy sources. High-quality prediction of electricity production has also direct economic consequences in estimating spot electricity prices and in planning energy supplies to the distribution network.

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